

Octavio Villalobos

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EDUCATION

University of California, Santa Cruz <i>Master of Science in Computer Science — GPA: 3.74/4.00</i>	June 2026 Santa Cruz, CA
University of California, Santa Cruz <i>Bachelor of Science in Computer Science, With Honors — GPA: 3.60/4.00</i>	June 2024 Santa Cruz, CA

EXPERIENCE

Research Project Lead @ Sloth Lab – UCSC <i>Santa Cruz, CA</i>	Oct 2025 – Present
– Led a 3-person team extending the Lyra opinion dynamics framework into a scalable multi-agent simulation modeling personality-driven NPC behavior across 1,000+ simulation runs – Designed agent architectures with 23-dimensional state tracking and per-agent Actor-Critic (A2C) RL policies across 5 personality archetypes using prospect-theory-based persuasion modeling – Built MLflow-tracked clustering and behavioral-analysis pipelines to validate emergent social behavior, culminating to a publication to AIIDE 2026	
AI Engineer Intern @ Silicon Valley Certification Hub <i>Palo Alto, CA</i>	June 2025 – Oct 2025, June 2026 - Present
– Owned end-to-end development of an LLM-powered coaching platform (Flask, AWS) built around a RAG pipeline, generating executive-style scenario assessments across 12 industries – Fine-tuned Gemma-3-1B and LLaMA-3.2-1B via LoRA (PEFT), improving factual alignment up to 25% and response consistency up to 9% on held-out evaluation data – Engineered rubric-based scoring pipelines using structured prompt engineering and ElevenLabs text-to-speech integration to evaluate relevance, risk coverage, and actionability	
Teaching Assistant @ University of California, Santa Cruz <i>Santa Cruz, CA</i>	Sep 2024 – June 2026
– Delivered lectures on databases, AI, LLMs, and technical writing to 150+ students across 4 UCSC courses – Led weekly discussion sections for 30 students, developing instructional materials and hands-on lesson plans – Designed coursework, assignments, and exams across 6 courses to improve lecture-assignment alignment	

PROJECTS

FormulaForecast <i>Python, PyTorch, scikit-learn</i>	Nov 2025 – Present
– Built a PyTorch deep learning framework predicting F1 top-10 finishing positions from FastF1 telemetry, engineering 9 domain-specific features with 3-sigma outlier clipping – Trained a 3-layer MLP (128→64→32) with Dropout and Batch Normalization, achieving 88.7% within-3 accuracy and cutting prediction error 41% versus baseline models	
Insight Deck <i>Python, React, Flask, LangChain, ChromaDB</i>	Jan 2025 – Jan 2026
– Co-developed a multi-agent RAG pipeline (LangChain) combining BM25 and ChromaDB vector search to automate investor presentation generation from financial filings – Engineered dynamic query-generation and LLM critique/repair pipelines, boosting content population from 74% to 98% and cutting slide-planning latency by 49%	
LLMs are ACDs <i>Python, CybORG, OpenAI API, HuggingFace</i>	Jan 2025 – June 2025
– Co-authored an IEEE CAI 2025 paper building the first LLM+RL agent framework for CybORG CAGE 4 multi-agent cyber defense, published as arXiv:2505.04843 – Built an 8-bit inter-agent communication protocol and Python API adapters (OpenAI, DeepSeek, HuggingFace), achieving ~104× faster action selection than RL baselines	

SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL (PostgreSQL, MySQL, SQLite) C#, HTML/CSS
Fullstack React.js, React Native (Expo), HTML5, Express.js, Node.js, RESTful APIs, Supabase
AI/ML & Generative AI: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, Hugging Face, OpenAI API, Claude API, LoRA, QLoRA, Ollama, ChromaDB, MLflow, Weights & Biases (W&B), Pandas, Matplotlib
Cloud & Infrastructure: AWS, Docker, Linux
DevOps & Testing: Git, GitHub Actions, CI/CD Pipelines, Pytest